FLIGHT OPERATIONS MANUAL / V2.1

OUTPOST SELENE

Lunar Survival Operations — Quick Start, Power Management, Crew Safety & Mission Controls

04 CREW MEMBERS	**48 kWh** BATTERY CAPACITY	**24 kWh/day** IDEAL SOLAR MAXIMUM
7 / 14 / 30 days MISSION LENGTHS	**T–30h** FIRST DUST-FRONT EVENT	**Separate hazards** DUST + SOLAR PARTICLES

YOUR MISSION

Keep all four crew members alive by balancing energy generation, life support, food production, equipment protection and time. Every time jump advances resource consumption and hazard timelines. The game teaches its core systems through visible feedback as well as this guide.

> **FIRST RULE: do not advance time blindly. Deploy and activate the solar array, start oxygen generation and water recycling, then inspect both hazard forecasts.**

IMPORTANT

This is an educational game model. Resource rates and hazard consequences are balancing rules, not real lunar mission specifications. The dust-front event is fictionalized for gameplay; the Moon does not have an atmosphere that creates Earth-style weather dust storms.

01 / FIRST SHIFT

Complete these setup tasks before spending time on optional activities.

STEP	ACTION	WHY IT MATTERS
1	Engineering → Deploy solar array	Panels start stowed. Deployment unfolds hardware but does not generate power.
2	Engineering → Activate solar generation	Generation requires daylight, deployed and active panels. Dust, damage and hazards reduce output.
3	Life Support → Start oxygen generator	Replenishes stored oxygen. If oxygen reaches zero, crew health falls rapidly.
4	Life Support → Start water recycler	Recovers part of daily water use. Water is still consumed continuously.
5	Engineering → Inspect storm shield	Reveals shield operating draw. This equipment shield is for the fictional dust-front gameplay event.
6	Weather → Track both timelines	Dust-front and solar-particle hazards differ. Prepare for each separately.

02 / POWER MANAGEMENT

Power is measured in kilowatt-hours (kWh). The battery stores up to 48 kWh. Solar collection is the only generation source in the game; active systems draw from that shared reserve.

POWER ITEM	GAME VALUE	HOW TO USE IT
Battery capacity	48 kWh maximum	Stored energy cannot exceed the battery cap. At capacity, extra generation is wasted.
Solar collection	Up to 2 kWh per daylight hour	12 daylight hours × 2 kWh = 24 kWh/day on an ideal clear day.
Essential daily load	About 18 kWh/day	Baseline outpost load + oxygen generator + water recycler.
Shield operating draw	0.45 kWh/hour	An 18-hour hazard window costs 8.1 kWh for the shield alone.
Recommended storm reserve	22 kWh	Target this before the dust-front event to cover essential loads plus shield use.

READ THE LIVE POWER PLAN: “DAILY IN / OUT” estimates solar energy collected versus estimated consumption. Battery storage is automatic and caps at 48 kWh. A negative balance means the reserve is shrinking; a positive balance means it can recharge.

03 / HAZARD RESPONSE

Two hazards appear in the mission. They have different causes, effects and responses.

A. DUST-FRONT GAMEPLAY EVENT

This fictionalized lunar gameplay challenge reduces solar input by about 86%, adds dust to panels and may damage exposed equipment. It is not a literal lunar atmospheric storm.

WARNING	PLAYER RESPONSE
T–24 hours	Open Weather. Inspect the shield, estimate energy needs, and secure exterior equipment.
T–12 hours	Check battery reserve, pause optional systems if needed, and recall the rover.
T–6 hours	Raise the equipment shield. Aim for roughly 22 kWh reserve.
T–1 hour	Final check: the shield MUST be ACTIVE. If it is OFF at impact, the mission fails immediately.
After impact	If protected, monitor power through the 18-hour event. Afterwards clean panel dust and repair damaged hardware.

B. SOLAR PARTICLE / RADIATION EVENT

This is a separate space-weather hazard representing energetic particles from the Sun. It is not the dust-front event. The equipment shield does not replace radiation protection for the crew.

WARNING	PLAYER RESPONSE
T–24 hours	Open Crew Quarters or Weather and identify the radiation countdown.
T–12 / 6 / 1 hour	Use Crew Quarters → Secure Radiation Shelter. Do not wait until the event begins.
During event	Keep crew sheltered for the six-hour event. Unsheltered crew accumulate exposure and lose health.
After event	Review crew health and exposure, then release the crew from shelter.

QUICK DISTINCTION: the dust-front hazard affects panel generation and hardware. The solar-particle event affects crew health. The equipment shield must be active at dust-front impact; the crew must be sheltered during radiation.

04 / ROOMS & GAMEPLAY LOOP

ROOM	PURPOSE / IMPORTANT ACTIONS
Engineering	Deploy/activate panels, inspect/raise/lower equipment shield, clean dust, repair hardware and watch the power budget.
Life Support	Start/stop oxygen generator and water recycler. Use emergency oxygen if needed.
Greenhouse	Start crop growth, monitor water/power, and manually harvest after 48 hours for +8 food rations.
Crew Quarters	Rest, use medical supplies, prepare exterior equipment, and secure the radiation shelter.
Rover Bay	Launch a three-hour survey for 6 kWh; successful return yields 2 materials and 1 research sample. Avoid hazards.
Research Lab	Analyze samples or update the forecast. Research improves information but should not outrank survival.
Weather	Monitor dust-front ETA, radiation ETA, warning milestones, panel risk and battery readiness.
Objectives	Review priorities and mission completion status.

05 / CONTROLS

CONTROL	WHAT IT DOES
Left sidebar	Switches room. Each room has its own camera focus and live telemetry.
Right operations rail	Shows available actions for the selected room.
+1H / +3H / +6H	Advances simulation. Large jumps pause at hazard warning milestones.
Keys 1–6	Quick-select Engineering, Life Support, Greenhouse, Crew Quarters, Rover Bay and Research Lab.
Space	Advance one hour (unless focus is in a text field).
Inspect / Field Guide	Opens briefing, room guide, weather, crew, mission log, objectives and settings.

06 / GUEST, SIGN-IN & SAVING

Guest: plays immediately with no saved progress. Save/Load are disabled and guest progress is not written to browser storage.

Create Account / Sign In: this build has a local-browser demo account flow. Account metadata and manual mission saves stay on the same browser/device; there is no cloud sync or server-backed authentication yet. Do not use this demo for sensitive passwords or production accounts. A production release needs a real identity provider and server-side access controls.

Save / Load: manual and account-only. Save before returning to the main menu if you want to preserve the current mission.

SURVIVAL CHECK: solar generation online · oxygen online · water recycling online · battery reserve tracked · rover secured before hazards · shield active at dust-front impact · crew sheltered during radiation.